

Applied Engineering Optimization

ENEL445 S1 2026

15 te hua | 15 points

Course co-ordinator:

Jeremy Watson

Īmēra/Waea | Email/phone contact details
jeremy.watson@canterbury.ac.nz, 03 3691810

Lecturer:

Le Yang

Īmēra/Waea | Email/phone contact details
le.yang@canterbury.ac.nz, 03 3690658

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Whakamahuki | Course Description

An advanced course on optimisation techniques and their engineering applications. The course first provides a review of the mathematical background, and then covers the formulation of unconstrained and constrained optimisation problems, gradient descent, method of Lagrangian multipliers, and first-order KKT conditions. Other topics include model predictive control, power system optimisation, evolutionary algorithms, constraint satisfaction methods, Bayesian optimization, nonlinear least squares, adaptive filtering and backpropagation for deep neural networks.

Optimisation is at the core of many engineering disciplines, including power systems, signal processing, communications, control, and machine learning. This course will cover the theory as well as the practice of engineering optimisation, including the problem formulation, problem transformation, gradient-based and gradient-free optimisation algorithms, and nature-inspired methods. Besides in-class lectures, students will work on a project which will give them hands-on experience solving real-world optimization problems. Topics include:

1. Mathematical background and optimisation basics
2. Gradient-based unconstrained optimisation
3. Constraint optimisation and KKT conditions
4. Optimisation-based control
5. Power system optimisation
6. Nature-inspired optimisation
7. FIR filter design using optimisation
8. Majorization-minorization (MM) algorithms
9. Deep neural networks

Hua Ako | Course Learning Outcomes

At the conclusion of this course, you should be able to:

- LO1: Identify, research, and formulate optimisation problems in power systems, signal processing, machine learning, and control (WA1, WA2, WA4)
- LO2: Classify the formulated optimisation problems, design and derive practical algorithms to search for the optimal solution (WA3, WA4, WA5)
- LO3: Using optimisation software to solve formulated optimisation problems, analyse the convergence of the algorithms and interpret the results (WA3, WA4, WA5)
- LO4: Collaborate with other students and lecturers to formulate a real-world engineering problem as an optimisation problem and design a solution using optimisation software (WA3, WA4, WA9, WA10, WA11)

Mahi ā-Ākonga | Workload

(expected distribution of student hours, note 15 points = 150 hours):

Activity	Time (hours)
Contact Hours	
Lectures	30
Tutorials	2
Workshops	0
Laboratories	0
Independent study	
Review of lectures	34

Test and exam preparation	24
Assignments	60
Tutorial preparation	0
Total	150

Aromatawai | Assessment (method, weight, date due)

Assessment	% of final grade	Date	Course Learning Outcomes	UC Graduate Profile	Washington Accord
Test	20	April 24	LO1-LO2	EIE	WA1, WA2, WA4
Progress Report *	10	March 20	LO1 – LO4	WA1 – WA5	EIE
Project Presentation	10	May 25	LO1 – LO4	WA8 – WA9	EIE
Final Report *	30	May 29	LO1 – LO4	WA1 – WA5, WA8 – WA10	EIE
Exam	30	June	LO1 – LO2	WA1, WA2, WA4	EIE
Total	100				

The examiners will award a failing grade to students who score less than 40% for the Test and Exam combined. More formally, $(\text{TestPercent} * 0.2 + \text{ExamPercent} * 0.3) / 0.5$ must be greater than or equal to 40 for a pass mark to be awarded. This is to ensure that each student has adequately shown to the examiners that they have gained some mastery of the topic.

Test 1 (20%)

On Friday 24 April 2026 (Week 1 of Term 2), a test worth 20% of the final grade will take place. The Test is closed book. UC stickered calculators are permitted. The test will cover Lecture material from Weeks 1-6 (Term 1) only.

Project (50%)

On Friday 18 February 2026 (Week 1 of Term 1), project options will be presented at the regular lecture time. There are two project options: 1) Source geo-location (run by Dr Le Yang); 2) Power systems optimization (run by Dr Jeremy Watson). Students must choose one of these two projects unless given approval for an alternative project by the course coordinator or lecturer. A breakdown of the assessment items associated with the project are:

Progress report (10%)

On Friday 20 March 2026 (Week 5 of Term 1), students must submit a progress report on Learn (project progress report dropdown). By departmental decision the deadline for all assignments and reports will be at 5:00pm on the day they fall due.

Final report (30%)

On Friday 29 May 2026 (Week 6 of Term 2), students must submit a progress report on Learn (project final report dropdown). By departmental decision the deadline for all assignments and reports will be at 5:00pm on the day they fall due.

Presentation (10%)

In Week 6 of Term 2, students will give presentations on their projects to the class and lecturers, followed by a short time for questions and answers.

Exam (30%)

In the exam period, the Exam worth 30% of the final grade will take place. The Exam is closed book. UC stickered calculators are permitted. The exam will mainly cover material taught in Weeks 7-12 (see indicative lecture plan below).

Lateness Penalties

For the Project progress and final reports, a lateness penalty of 10% (in absolute terms) per day or part day late will be deducted from the original mark. For example, an assignment with a nominal mark of 83% submitted 0-24 hours late will receive a mark of 73%, and submitted 24-48 hours late will receive 63%.

Course Breakdown

Week beginning	Lecture material	Assessment
16 February	Introduction, mathematical background, projects	
23 February	Unconstrained optimisation, gradient-based descent, numerical computation of gradient	
2 March	Review of least squares, ridge regression, LASSO, majorization-minorization	
9 March	Constrained optimisation and method of Lagrangian multiplier	
16 March	Optimal control / MPC / dynamic programming	Project progress report (10%)
23 March	Gradient-free optimisation	

Week beginning	Lecture material	Assessment
20 April	Reinforcement learning	Test (20%)
27 April	Power systems optimisation	
4 May	Alternating Direction Method of Multipliers (ADMM).	
11 May	Linear Matrix Inequalities (LMIs)	
18 May	Game theory, k-means clustering, Gaussian mixture models	
25 May	(Project presentations)	Project final report (30%) Project presentation (10%)
Exam period		Exam (30%)

Notes:

1. Weeks 9-10 are planned to be taught by the Erskine visitor, Prof. Roy Smith, ETH Zurich.
2. The lecture content is subject to change in the case of unforeseen circumstances (e.g. due to medical matters affecting one of the lecturers). However, we will endeavour to stick with the plan above.

Tuhinga | Texts and Readings

Required Textbook:

<u>Authors</u>	<u>Title</u>	<u>Publisher</u>
J. Martins and A. Ning	Engineering Design Optimization	Cambridge University Press, 2022. Freely available from https://mdobook.github.io/

Other texts:

Authors	Title	Publisher
Z. Michalewicz and D. B. Fogel	How to solve it: Modern Heuristics	Springer 1998
R. Baldick	Applied Optimization: Formulation and Algorithms for Engineering Systems	Cambridge University Press, 2006
J. Nocedal and S. J. Wright	Numerical Optimization	Springer, 2000
A. Beck	Introduction to Nonlinear Optimziation	SIAM, 2014.
S. Brunton	Optimization Bootcamp	OptimizationBootcamp.pdf

Herenga Akoranga | Academic Policies (e.g. special consideration, dishonest practice)

Scaling of marks

In order to maintain consistency across courses and fairness for students, scaling of raw marks occurs. In the Faculty of Engineering, target course GPAs are calculated based on the performance of the cohort of students in their courses in the previous year. Scaling of the raw total course marks is normally performed so that when converted to grades (using UC Grade Scale) the outgoing GPA is in line with the target GPA for a course. Scaling up or down can occur.

The Grading Scale for the University:

<https://www.canterbury.ac.nz/study/study-support-info/study-related-topics/grading-scale>

Artificial Intelligence Tools

The use of Artificial Intelligence (AI) tools for each of the assessments in ENEL445 is summarised in the Table below. No AI use is allowed in the tests and exam because these are closed-book invigilated assessments. Students are always responsible for the accuracy of the submitted works, regardless of which tools are used.

Assessment Item	Permitted use of AI	
Project presentation	Generative AI Tools Are Permitted for Certain Parts of This Assessment	
Project reports	Generative AI Tools Are Permitted for Certain Parts of This Assessment	
Test	Generative AI tools cannot be used for this assessment.	
Exam	Generative AI tools cannot be used for this assessment.	

Generative AI Tools Are Permitted for Certain Parts of This Assessment

In the project-based assessments, you are permitted to use generative artificial intelligence (AI) for the purpose of proof reading and editing documents, for debugging code, and for gathering and summarising knowledge. No other use of generative AI is permitted, including the generation of substantial parts of your computer code or report. To assist with maintaining academic integrity, you must appropriately acknowledge any use of generative AI in your work. Please include a Statement of AI use (if no AI tool has been used, then this must also be stated) and a listing of all prompts provided to the AI tool, clearly indicating which AI tools were used and how they contributed to your assessment.

Special Considerations

As per the Faculty of Engineering policy, the Professional years of the BE(Hons) degree no longer use computed grades as a remedy for special considerations.

For the invigilated assessments (the Tests and Exam), the default remedy will be a delayed or repeated sitting of an alternative assessment item that shows equivalent learning outcomes; if a delayed or repeated sitting is not appropriate and evidence shows that the course learning outcomes are already met, the student may be awarded a pass grade (P AEG).

In the case of a re-sit of either the Tests or the Exam, the grade in the second attempt only is used to calculate the final grade. The Exam resit will take place in the Week of Monday 6 July – Thursday 9 July. Students should inform the Course Coordinator if they have applied for Special Consideration for the Tests or Exam.

For the Project Progress and Final Reports, the default remedy will be an extension to the submission deadline. Students should apply directly to the Course Co-ordinator if they meet the Special Considerations criteria (with supporting evidence e.g. medical certificate) rather than through the Special Consideration process. Extensions

cannot be granted for having several assessments due at the same time, mistaking the due date or time, or other work commitments. For further information on Special Considerations, refer to: <https://www.canterbury.ac.nz/study/special-consideration/>.

Assessment Completion

Students must complete and submit the Project Progress and Final Report in order to be eligible to pass the course as defined in the General Conditions for Credit Regulations 3(a) [here](#).

Assessment and Grade Appeals

Within five working days of receiving the results of, or feedback for, an assessment (whichever is later), a student may apply to the Course Coordinator to have the assessment reconsidered. [Taught and Project Course Assessment Regulation 7(a) [here](#)]

Within four weeks of the final course grades being released, a student may apply for reconsideration of a final grade on the appropriate form. Only assessment items that have not already been returned to students will be fully re-marked, while the marks of previous assessments will be checked. Note that a fee is charged, but this is refunded if a change of grade occurs. [Taught and Project Course Assessment Regulation 7(d) [here](#)]

For further information on a reconsideration of grade, see <https://www.canterbury.ac.nz/study/study-support-info/study-related-topics/examinations/exam-results-dates-and-appeals>

Emergency Provisions

In the case of an emergency that affects the whole course, the Course Coordinator, in consultation with the Dean, may change the nature, weighting and timing of assessments, e.g. tests and examination may be replaced with assignments of the same weight or different weight at a different time and/or date (which, under certain circumstances, may be outside the prescribed course dates). The 'Special consideration' process will also be used for unforeseen circumstances that adversely affect the academic performance of students individually. The usual grounds for this are described in the UC policy 'Special Consideration Procedures and Guidelines', and personal circumstances due to a wider emergency event may also qualify

Dishonest Practice

Plagiarism, collusion, copying and ghost writing are unacceptable and dishonest practices.

- Plagiarism is the presentation of any material (text, data, figures or drawings, on any medium including computer files) from any other source without clear and adequate acknowledgment of the source.
- Collusion is the presentation of work performed in conjunction with another person or persons, but submitted as if it has been completed only by the named author(s).
- Copying is the use of material (in any medium, including computer files) produced by another person(s) with or without their knowledge and approval.
- Ghost writing is the use of another person(s) (with or without payment) to prepare all or part of an item submitted for assessment.

Do not engage in dishonest practices. The Department reserves the right to refer dishonest practices to the University Proctor and where appropriate to not mark the work.

Harassment of any sort will not be tolerated

Each student is here at UC to learn and to experience a friendly and supportive community. It is every student's right to expect: respect and courtesy from staff and other students, including freedom from harassment of any sort; fair treatment; the ability to speak out about any issues that concern them, without fear of consequences

for their safety and well-being. Furthermore, each student has the responsibility to: respect the rights and property of others; attend to their own health and safety and that of others; and behave in a manner towards each other that does not reflect badly on the student body or the University.

For more information, or to find out where help is available, refer to:
www.canterbury.ac.nz/ucpolicy/GetPolicy.aspx?file=Harassment-Policy.pdf.

UC Graduate Profile

EIE	Employable, Innovative & Enterprising • Working effectively and professionally with diverse communities • Communication • Analytical, critical thinking and problem solving in diverse contexts • Digital literacy • Innovation, enterprising and creativity	x
BCC	Bicultural Competence & Confidence • A process of self-reflection on the nature of 'knowledge' and 'norms' • The nature of contemporary Māori organisational structures, e.g. rūnanga, hapū, iwi, iwi corporations • Traditional and contemporary realities of Māori society, e.g. tikanga and kawa, te reo Māori • The Treaty of Waitangi and Aotearoa New Zealand's bicultural history • The processes of colonisation and globalisation • Other Indigenous models of development, knowledge and behaviours • Application of bicultural competence and confidence in a chosen discipline and career	
GA	Globally Aware • Self-reflection on the nature of one's culture, language and beliefs and on one's systems of knowledge • Understanding the global nature of one's discipline • The ability to engage critically and effectively in global and multicultural contexts	
CE	Community Engagement • Engagement: Gaining knowledge and understanding of a community by interacting with a community • Reflection: Gaining knowledge and understanding of a community through reflection on one's experience with that community • Application: Understanding and articulating how the content and/or skills of the subject/programme enhances the community	

Washington Accord

Blank (none)

*Underpinning knowledge is provided

**Basic learning occurs and is assessed

***More advanced learning occurs

****More advanced learning occurs and is robustly assessed

Washington Accord Outcomes		Level
WA1	Engineering knowledge	***
	Apply knowledge of mathematics, natural science, computing and engineering fundamentals, and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions to complex engineering problems.	
WA2	Problem analysis	***
	Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development* (WK1 to WK4). *	
WA3	Design / development of solutions	*
	WA3: Design creative solutions for complex engineering problems and design systems, components or processes to meet identified needs with appropriate consideration for public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal, and environmental considerations as required (WK5)	
WA4	Investigation	***
	Conduct investigations of complex engineering problems using research methods including research based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions (WK8)	
WA5	Tool usage	***
	Create, select and apply, and recognize limitations of appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems (WK2 and WK6)	
WA6	The Engineer and The World	
	When solving complex engineering problems, analyze and evaluate sustainable development impact (Represented by the 17 UN Sustainable Development Goals (UN-SDG) to: society, the economy, sustainability, health and safety, legal frameworks, and the environment (WK1, WK5, and WK7)	
WA7	Ethics	
	Apply ethical principles and commit to professional ethics and norms of engineering practice and adhere to relevant national and international laws. Demonstrate an understanding of the need for diversity and inclusion (WK9)	
WA8	Individual and Collaborative Team work	*
	Function effectively as an individual, and as a member or leader in diverse and inclusive teams and in multi-disciplinary, face-to-face, remote and distributed settings (WK9)	

Blank (none)

*Underpinning knowledge is provided

**Basic learning occurs and is assessed

***More advanced learning occurs

****More advanced learning occurs and is robustly assessed

Washington Accord Outcomes		Level
WA9	Communication	**
	Communicate effectively and inclusively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, taking into account cultural, language, and learning differences	
WA10	Project management and finance	**
	Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.	
WA11	Lifelong learning	
	Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change (WK8)	

Washington Accord Knowledge Profile

WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.

WK2: Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.

WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.

WK5: Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.

WK6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

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WK7: Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.

WK8: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

WK9: Ethics, inclusive behaviour and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.